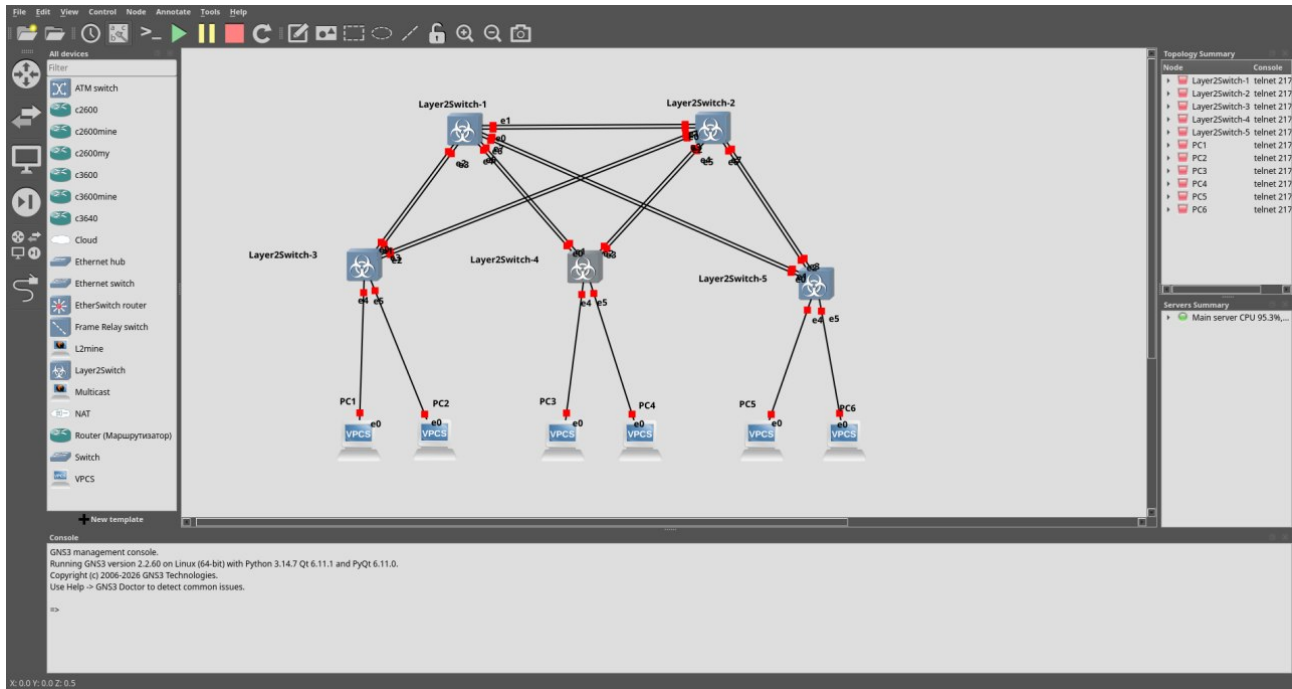


1) Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров

настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета



Выберем Switch-4 как корневой, задав ему наименьший приоритет

Команды:

enable

configure terminal

spanning-tree vlan 1 priority 4096

end

write

```
vIOS-L2-01#enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree vlan 1 priority 4096
vIOS-L2-01(config)#end
vIOS-L2-01#write
*Sep 1 08:37:03.962: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#write
Building configuration...

-Traceback= 1C489D8z 8BD015z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77
-Traceback= 1C489D8z 8BD015z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 889E8Dz 889D8Dz 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889FE2
Compressed configuration from 5070 bytes to 1945 bytes
*Sep 1 08:37:07.892: %SYS-3-CPUHOG: Task is running for (2000)msecs, more than (2000)msecs (30/8),process = Async write process.
-Traceback= 1C489D8z 8BD015z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77
-Traceback= 1C489D8z 8BD015z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77
-Traceback= 1C489D8z 8BD015z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77
-Traceback= 1C489D8z 8BD015z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77
-Traceback= 1C489D8z 8BD015z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77
[OK]
*Sep 1 08:37:11.844: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than (2000)msecs (34/19),process = Async write process.
*Sep 1 08:37:14.364: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than (2000)msecs (68/40),process = Async write process.
*Sep 1 08:37:16.728: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than (2000)msecs (30/9),process = Async write process.
*Sep 1 08:37:18.976: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than (2000)msecs (50/30),process = Async write process.
*Sep 1 08:37:23.280: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than (2000)msecs (51/28),process = Async write process.
*Sep 1 08:37:23.487: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
vIOS-L2-01#
*Sep 1 08:37:24.992: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
vIOS-L2-01#
```

На остальных коммутаторах назначим больший приоритет:

```
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree vlan 1 priority 8192
vIOS-L2-01(config)#end
vIOS-L2-01#wri
*Sep  1 08:39:47.932: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#write
Building configuration...

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B2
83z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC
0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B2
83z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC
0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B2
83z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC
0x00889D77
Compressed configuration from 5078 bytes to 1946 bytes
*Sep  1 08:39:59.270: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than (
2000)msecs (38/16),process = Async write process.
*Sep  1 08:40:01.886: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than (
2000)msecs (61/40),process = Async write process.
-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B2
83z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC
0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 889E8Dz 889D
8Dz 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write proces
s", CPU hog, PC 0x00889FE2
[OK]
*Sep  1 08:40:03.982: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than (
2000)msecs (18/0),process = Async write process.
*Sep  1 08:40:06.790: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than (
2000)msecs (44/25),process = Async write process.
*Sep  1 08:40:09.706: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than (
2000)msecs (13/0),process = Async write process.
*Sep  1 08:40:14.418: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated
on disk. Please wait...
vIOS-L2-01#
*Sep  1 08:40:15.748: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to d
isk successfully.
vIOS-L2-01#
```

2) Проверить доступность каждого с каждым всех персональных компьютеров (VPCS),
результаты запротоколировать

зададим узлам ip адреса

```
PC1> ip 192.168.1.1/24 192.168.1.254
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0 gateway 192.168.1.254

PC1> save
Saving startup configuration to startup.vpc
. done
```

```
PC2> ip 192.168.1.2/24 192.168.1.254
Checking for duplicate address...
PC2 : 192.168.1.2 255.255.255.0 gateway 192.168.1.254

PC2> save
Saving startup configuration to startup.vpc
. done
```

```
PC3> ip 192.168.1.3/24 192.168.1.254
Checking for duplicate address...
PC3 : 192.168.1.3 255.255.255.0 gateway 192.168.1.254

PC3> save
Saving startup configuration to startup.vpc
. done
```

```
PC4> ip 192.168.1.4/24 192.168.1.254
Checking for duplicate address...
PC4 : 192.168.1.4 255.255.255.0 gateway 192.168.1.254

PC4> save
Saving startup configuration to startup.vpc
. done
```

```
PC5> ip 192.168.1.5/24 192.168.1.254
Checking for duplicate address...
PC5 : 192.168.1.5 255.255.255.0 gateway 192.168.1.254

PC5> save
Saving startup configuration to startup.vpc
. done
```

```
PC6> ip 192.168.1.6/24 192.168.1.254
Checking for duplicate address...
PC6 : 192.168.1.6 255.255.255.0 gateway 192.168.1.254

PC6> save
Saving startup configuration to startup.vpc
. done
```

Выполним Ping для каждой пары:

1 к 2

```
PC1> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=12.461 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=7.475 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=6.695 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=13.936 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=18.605 ms
```

1 к 3

```
PC1> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=24.350 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=29.712 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=20.092 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=16.420 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=46.214 ms
```

1 к 4

```
PC1> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=21.364 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=26.343 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=40.289 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=18.256 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=32.794 ms
```

1 к 5

```
PC1> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=33.727 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=18.179 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=20.441 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=34.564 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=15.490 ms
```

1 к 6

```
PC1> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=55.351 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=31.505 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=18.164 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=25.441 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=37.315 ms
```

2 к 3 4 5 6

```
PC2> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=13.807 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=19.707 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=34.526 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=26.374 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=26.590 ms

PC2> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=49.871 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=21.457 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=39.325 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=19.248 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=31.366 ms

PC2> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=17.028 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=32.643 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=26.197 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=28.989 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=42.425 ms

PC2> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=23.286 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=21.340 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=25.297 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=37.480 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=29.497 ms
```

3 к 4 5 6

```
PC3> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=13.863 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=9.683 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=7.848 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=21.273 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=6.923 ms

PC3> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=18.085 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=27.870 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=19.243 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=10.385 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=41.629 ms

PC3> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=20.032 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=29.079 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=35.358 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=35.009 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=25.118 ms
```


4 к 5 6

```
PC4> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=31.305 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=23.123 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=19.398 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=26.480 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=38.483 ms

PC4> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=52.208 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=18.621 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=19.291 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=11.581 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=21.319 ms
```

5 к 6

```
PC5> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=11.940 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=6.816 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=2.210 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=9.069 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=5.913 ms
```

3) На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл

SW1

```
vIOS-L2-01>show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
             Address     0c1f.c69b.0000
             Cost        4
             Port        5 (GigabitEthernet1/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Dec

  Bridge ID   Priority    8193 (priority 8192 sys-id-ext 1)
             Address     0cfb.a275.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Dec
             Aging Time   300 sec

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/0          Altn BLK 4         128.1   Shr
Gi0/1          Altn BLK 4         128.2   Shr
Gi0/2          Desg FWD 4         128.3   Shr
Gi0/3          Desg FWD 4         128.4   Shr
Gi1/0          Root FWD 4         128.5   Shr
Gi1/1          Altn BLK 4         128.6   Shr
Gi1/2          Desg FWD 4         128.7   Shr
Gi1/3          Desg FWD 4         128.8   Shr
Gi2/0          Desg FWD 4         128.9   Shr
```

SW2

```
vIOS-L2-01>show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
             Address     0c1f.c69b.0000
             Cost        4
             Port        5 (GigabitEthernet1/0)
             Hello Time   2 sec  Max Age 20 sec  Forwardc

  Bridge ID   Priority    8193  (priority 8192 sys-id-ex)
             Address     0c10.8d90.0000
             Hello Time   2 sec  Max Age 20 sec  Forwardc
             Aging Time   300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Desg FWD 4         128.1   Shr
Gi0/1                    Desg FWD 4         128.2   Shr
Gi0/2                    Desg FWD 4         128.3   Shr
Gi0/3                    Desg FWD 4         128.4   Shr
Gi1/0                    Root FWD 4         128.5   Shr
Gi1/1                    Altn BLK 4         128.6   Shr
Gi1/2                    Desg FWD 4         128.7   Shr
Gi1/3                    Desg FWD 4         128.8   Shr
Gi2/0                    Desg FWD 4         128.9   Shr
```

SW3

```
vIOS-L2-01>show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
             Address     0c1f.c69b.0000
             Cost        8
             Port        3 (GigabitEthernet0/2)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    8193  (priority 8192 sys-id-ext 1)
             Address     0c7b.cf54.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time   300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Altn BLK 4         128.1   Shr
Gi0/1                    Altn BLK 4         128.2   Shr
Gi0/2                    Root FWD 4         128.3   Shr
Gi0/3                    Altn BLK 4         128.4   Shr
Gi1/0                    Desg FWD 4         128.5   Shr
Gi1/1                    Desg FWD 4         128.6   Shr
```

SW4

```
vIOS-L2-01>show spanning-tree
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    4097
           Address    0c1f.c69b.0000
           This bridge is the root
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID  Priority    4097 (priority 4096 sys-id-ext 1)
           Address    0c1f.c69b.0000
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
           Aging Time 300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Desg	FWD	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr

SW5

```
vIOS-L2-01>show spanning-tree
```

```
VLAN0001
```

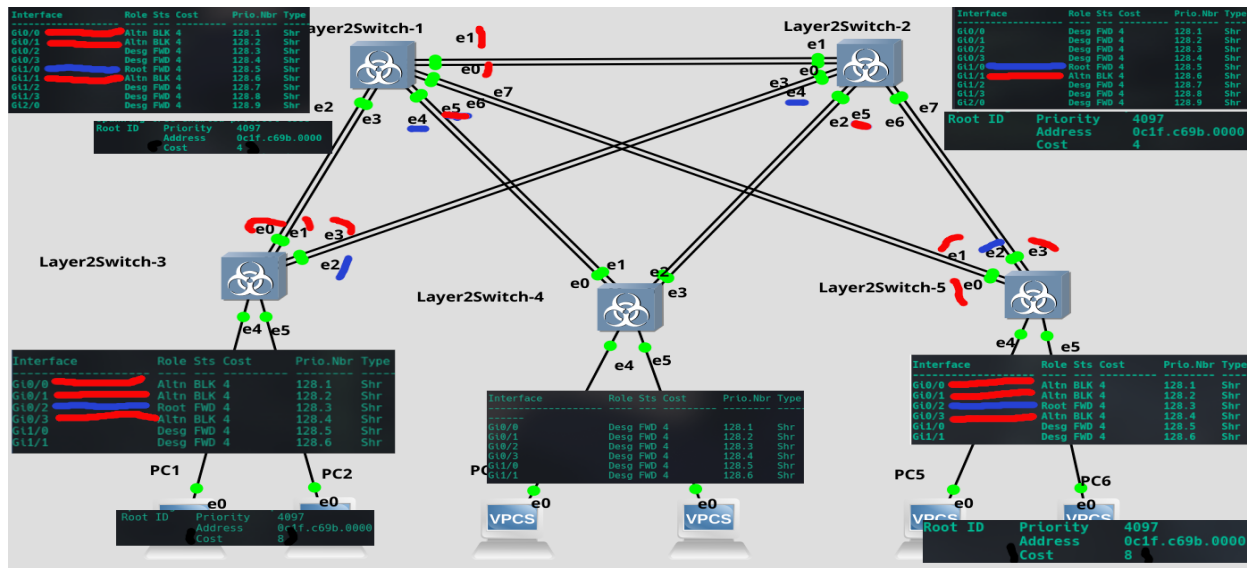
```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    4097
           Address    0c1f.c69b.0000
           Cost        8
           Port        3 (GigabitEthernet0/2)
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID  Priority    8193 (priority 8192 sys-id-ext 1)
           Address    0cbe.4388.0000
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
           Aging Time 300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Altn	BLK	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Root	FWD	4	128.3	Shr
Gi0/3	Altn	BLK	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr

Схема "lab2-scheme.png".(красный-заблокирован/альтернативный, синий - корневой)



4) При помощи wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет

Линк e0(SW4-SW1):

No.	Time	Delta	Source	Destination	Protocol	Length	Info
13	25.738891		0c:1f:c6:9b:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:00
16	28.158121	2.419230	0c:1f:c6:9b:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:00
20	31.849790	3.691669	0c:1f:c6:9b:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:00
24	34.889095	3.039305	0c:1f:c6:9b:00:00	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:1f:c6:9b:00:00
26	37.566583	2.677488	0c:1f:c6:9b:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 8192/1/0c:be:43:88:00:00
31	40.008628	2.442045	0c:fb:a2:75:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:1f:c6:9b:00:01
49	45.698179	5.689551	0c:1f:c6:9b:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:1f:c6:9b:00:01
50	45.699753	0.001574	0c:1f:c6:9b:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:1f:c6:9b:00:01
51	45.702098	0.002345	0c:1f:c6:9b:00:00	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:1f:c6:9b:00:00
52	45.707174	0.005076	0c:1f:c6:9b:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:1f:c6:9b:00:01
53	46.010841	0.303667	0c:fb:a2:75:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:fb:a2:75:00:05

Линк e1(SW4-SW1):

No.	Time	Delta	Source	Destination	Protocol	Length	Info
13	25.739026		0c:1f:c6:9b:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:00
16	28.162579	2.423553	0c:1f:c6:9b:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:00
20	31.849872	3.687293	0c:1f:c6:9b:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:00
24	34.888784	3.038912	0c:1f:c6:9b:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:1f:c6:9b:00:00
26	37.567048	2.678264	0c:1f:c6:9b:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 8192/1/0c:be:43:88:00:00
31	40.008659	2.441611	0c:fb:a2:75:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:1f:c6:9b:00:01
48	45.698307	5.689648	0c:1f:c6:9b:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:1f:c6:9b:00:01
49	45.700585	0.002278	0c:1f:c6:9b:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:1f:c6:9b:00:01
50	45.702183	0.001598	0c:1f:c6:9b:00:01	Nearest-Customer-Br...	STP	60	Conf. TC + Root = 4096/1/0c:1f:c6:9b:00:00
51	45.708220	0.006037	0c:1f:c6:9b:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:fb:a2:75:00:05
52	46.022958	0.314738	0c:fb:a2:75:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:fb:a2:75:00:05
55	46.869867	0.846909	0c:fb:a2:75:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:fb:a2:75:00:05
56	46.870916	0.001049	0c:fb:a2:75:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:fb:a2:75:00:05
57	46.872039	0.001123	0c:fb:a2:75:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:fb:a2:75:00:05

Линк e2(SW4-SW2):

No.	Time	Delta	Source	Destination	Protocol	Length	Info
52221	18397.699203	0.015402	0c:1f:c6:9b:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:02
52222	18398.158230	0.459027	0c:10:8d:90:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:10:8d:90:00:04
52223	18398.567654	0.409424	0c:10:8d:90:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:10:8d:90:00:04
52224	18399.165598	0.597944	0c:1f:c6:9b:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:02
52225	18400.198430	1.032832	0c:1f:c6:9b:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:1f:c6:9b:00:02
52226	18400.602982	0.404552	0c:1f:c6:9b:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:1f:c6:9b:00:02
52227	18400.729879	0.126897	0c:1f:c6:9b:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:1f:c6:9b:00:02
52228	18401.377240	0.647361	0c:10:8d:90:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:10:8d:90:00:04
52229	18401.383745	0.006505	0c:1f:c6:9b:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:02
52230	18401.900274	14293.040354	0c:10:8d:90:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:10:8d:90:00:04
52231	18402.271763	1241.628468	0c:10:8d:90:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:10:8d:90:00:04
52232	18402.856053	1.472308	0c:1f:c6:9b:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:02
52233	18403.130106	0.274053	0c:1f:c6:9b:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:1f:c6:9b:00:02

Линк e3(SW4-SW2):

No.	Time	Delta	Source	Destination	Protocol	Length	Info
52202	18449.433030	0.356026	0c:10:8d:90:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:10:8d:90:00:05
52203	18450.147227	0.714197	0c:1f:c6:9b:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:03
52204	18451.501534	1.354307	0c:1f:c6:9b:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:1f:c6:9b:00:03
52206	18451.931688	0.430154	0c:1f:c6:9b:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:1f:c6:9b:00:03
52207	18452.044231	0.112543	0c:1f:c6:9b:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:1f:c6:9b:00:03
52208	18452.085769	0.041538	0c:10:8d:90:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:10:8d:90:00:05
52209	18452.087815	0.002046	0c:1f:c6:9b:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:03
52210	18452.590497	0.502682	0c:10:8d:90:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:10:8d:90:00:05
52211	18452.983748	0.393251	0c:10:8d:90:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:10:8d:90:00:05
52212	18453.583183	0.599435	0c:1f:c6:9b:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:03
52213	18454.574867	0.991684	0c:1f:c6:9b:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:1f:c6:9b:00:03
52214	18454.960132	8515.771964	0c:1f:c6:9b:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:1f:c6:9b:00:03

Линк e4(SW4-PC3):

No.	Time	Delta	Source	Destination	Protocol	Length	Info
8291	18408.399284	2.984838	0c:1f:c6:9b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:04
8293	18411.301337	2.902053	0c:1f:c6:9b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:04
8294	18414.335846	3.034509	0c:1f:c6:9b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:04
8295	18417.442203	3.106357	0c:1f:c6:9b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:04
8298	18420.548262	3.106059	0c:1f:c6:9b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:04
8299	18423.559771	3.011509	0c:1f:c6:9b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:04
8301	18426.390773	2.831002	0c:1f:c6:9b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:04
8302	18429.489549	3.098776	0c:1f:c6:9b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:04
8303	18432.460612	2.971063	0c:1f:c6:9b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:04
8304	18435.536171	3.075559	0c:1f:c6:9b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:04
8305	18438.567859	3.031688	0c:1f:c6:9b:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:04

Линк e5(SW4-PC4):

No.	Time	Delta	Source	Destination	Protocol	Length	Info
8305	18438.568216	3.031935	0c:1f:c6:9b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:05
8307	18441.453873	2.885657	0c:1f:c6:9b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:05
8308	18444.178784	2.724911	0c:1f:c6:9b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:05
8309	18446.888750	2.709966	0c:1f:c6:9b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:05
8310	18449.858463	2.969713	0c:1f:c6:9b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:05
8311	18452.806557	2.948094	0c:1f:c6:9b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:05
8313	18455.853399	3.046842	0c:1f:c6:9b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:05
8314	18458.781747	2.928348	0c:1f:c6:9b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:05
8316	18461.743011	2.961264	0c:1f:c6:9b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:05
8319	18464.445733	2.702722	0c:1f:c6:9b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:05
8320	18467.376524	2.930791	0c:1f:c6:9b:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:1f:c6:9b:00:05

заголовки STP кадра полученного на линке e0:

▶ Frame 47664: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0	
▼ Ethernet II, Src: 0c:1f:c6:9b:00:00 (0c:1f:c6:9b:00:00), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)	0000 01 80 c2 00 00 00 0c 1f c6 9b 00 00 81 00 00 64
▼ Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)	0010 00 26 42 42 03 00 00 00 00 80 64 0c 1f c6 9b
.....0..... = LG bit: Globally unique address (factory default)	0020 00 00 00 00 00 00 80 64 0c 1f c6 9b 00 00 80 01
.....1..... = IG bit: Group address (multicast/broadcast)	0030 00 00 14 00 02 00 0f 00 00 00 00 00
▼ Source: 0c:1f:c6:9b:00:00 (0c:1f:c6:9b:00:00)	
.....0..... = LG bit: Globally unique address (factory default)	
.....0..... = IG bit: Individual address (unicast)	
Type: 802.1Q Virtual LAN (0x8100)	
[Stream index: 7]	
▼ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 100	
000..... = Priority: Best Effort (default) (0)	
...0..... = DEI: Ineligible	
....0000 0110 0100 = ID: 100	
Length: 38	
Trailer: 00000000	
▼ Logical-Link Control	
▶ DSAP: Spanning Tree BPDU (0x42)	
▶ SSAP: Spanning Tree BPDU (0x42)	
▶ Control field: U, func=UI (0x03)	
▼ Spanning Tree Protocol	
Protocol Identifier: Spanning Tree Protocol (0x0000)	
Protocol Version Identifier: Spanning Tree (0)	
BPDU Type: Configuration (0x00)	
▶ BPDU flags: 0x00	
▶ Root Identifier: 32768 / 100 / 0c:1f:c6:9b:00:00	
▶ Root Path Cost: 0	
▶ Bridge Identifier: 32768 / 100 / 0c:1f:c6:9b:00:00	
▶ Port identifier: 0x8001	
Message Age: 0	
Max Age: 20	
Hello Time: 2	
Forward Delay: 15	

Разберем заголовки STP кадра полученного на линке e0:

Заголовки:

Идентификатор протокола: Протокол остового дерева (Spanning Tree Protocol) 0x0000

Идентификатор версии протокола: Остовая сеть (Spanning Tree) 0

Тип BPDU: Конфигурация (Configuration) 0x00

Флаги BPDU: 0x00

Идентификатор корневого моста: 32768 / 100 / 0c:1f:c6:9b:00:00

Стоимость пути до корня: 0

Идентификатор моста: 32768 / 100 / 0c:1f:c6:9b:00:00

Идентификатор порта: 0x8001

Возраст сообщения: 0

Максимальный возраст: 20

Интервал Hello: 2

Задержка пересылки: 15

5) Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл

Для SW1:

configure terminal

interface GigabitEthernet1/0

spanning-tree cost 100

end

write

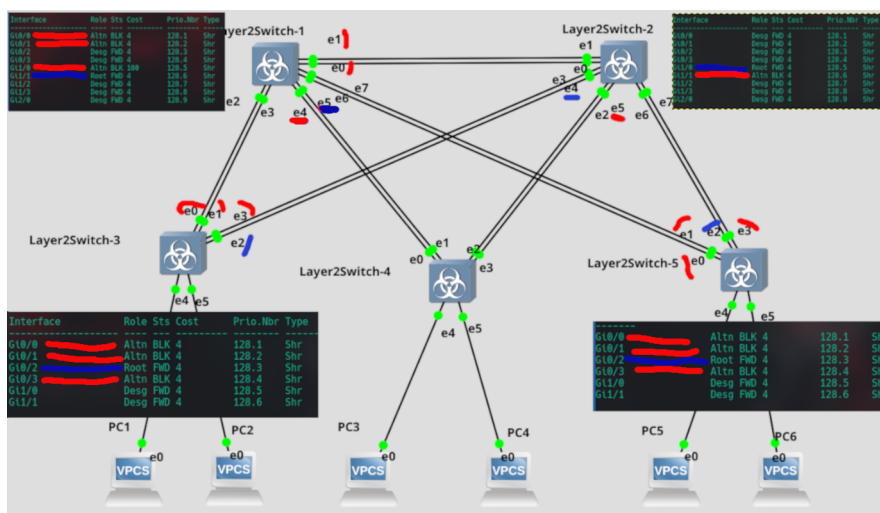
было

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Altn	BLK	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Root	FWD	4	128.5	Shr
Gi1/1	Altn	BLK	4	128.6	Shr
Gi1/2	Desg	FWD	4	128.7	Shr
Gi1/3	Desg	FWD	4	128.8	Shr
Gi2/0	Desg	FWD	4	128.9	Shr

стало

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Altn	BLK	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Altn	BLK	100	128.5	Shr
Gi1/1	Root	FWD	4	128.6	Shr
Gi1/2	Desg	FWD	4	128.7	Shr
Gi1/3	Desg	FWD	4	128.8	Shr
Gi2/0	Desg	FWD	4	128.9	Shr

схема lab2-scheme-2.png сохранена



б) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

Файлы конфигураций коммутаторов сохранены в директорию configs.